

controller, which can lead to missing writes or buggy writes, they cannot detect errors that originate in the file system code. So given the possibility that file systems can become inconsistent due to errors occurring either outside the FS in the storage stack or due to buggy code in the FS itself, how do we check the consistency of the file system—and if not consistent, bring it to a consistent state?

File system consistency checking

FSCK or file system consistency check is a utility that is traditionally used to perform a check on the consistency of the file system; if inconsistencies are found, it can repair them automatically or, in certain cases, with the help of the user. Windows users would know it by its avatar, 'chkdsk'.

File systems inconsistencies can arise due to an:

(a) unclean shutdown of the file system, typically due to either power failure or the user not following proper shutdown procedures; or (b) due to hardware failures, which lead to the file system meta-data information on the disk becoming inconsistent. Allowing a corrupted file system to be used can lead to further inconsistencies and, in certain cases, even to permanent data loss. Hence, when systems are brought back online after a crash, operators have been known to run the 'fsck' before the file system can be made online and user IO operations allowed on it.

FSCK independently tries to build its knowledge of the structure and layout of the file system from the various data structures on disk, and corroborates it with the summary/computed information maintained by the file system. If the two pieces of information don't match, an inconsistency is detected and FSCK tries to repair the inconsistency. If automatic repair is not possible, the problem is reported to the user. A good overview of FSCK can be found in <http://lwn.net/Articles/248180/>.

Here is a question to our readers. Do all file systems need a FSCK utility? For instance, there are file systems that support journaling or write-ahead logging, wherein all changes to metadata are first logged on to a journal log on persistent media before the metadata itself is updated. File system inconsistencies created by partial meta-data writes resulting from a sudden crash while operations are in mid-flight are addressed by means of the file system journal log, which can replay the log on restart and recover to a consistent state. Do such journaling file systems need FSCK?

My 'must-read book' for this month

This month's must-read book suggestion comes from one of our readers, Sonia George. She recommends 'In Search of Clusters' (2nd Edition) by Gregory F Pfister, which discusses the internals of cluster computing and provides details of different clustering implementations. Thanks, Sonia, for your suggestion.

If you have a favourite programming book or article that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful so that I can mention it in this column. This would help many readers who want to improve their software skills.

If you have any favourite programming questions or software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming and here's wishing you the very best! 

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